Efficient Fine-Grained Object Recognition in High-Resolution Remote Sensing Images From Knowledge Distillation to Filter Grafting

Liuqian Wang, Jing Zhang, *Member, IEEE*, Jimiao Tian, Jiafeng Li, Li Zhuo, and Qi Tian, *Fellow, IEEE*

***Abstract*— With the development of high-resolution remote sensing images (HR-RSIs) and the escalating demand for intelligent analysis, fine-grained recognition of geospatial objects has become a more practical and challenging task. Although deep learning-based object recognition has achieved superior performance, it is inflexible to be directly utilized to the fine-grained object recognition (FGOR) tasks of HR-RSIs under the limitation of the size of geospatial objects. An efficient fine-grained object recognition method in HR-RSIs from knowledge distillation (KL) to filter grafting is proposed. Specifically, fine-grained object recognition consists of two stages: Stage 1 utilizes oriented region convolutional neural network (oriented R-CNN) to accurately locate and preliminarily classify geospatial objects. At the same time, it serves as a teacher network to guide students' effective learning of fine-grained object recognition; in Stage 2, we design a coarse-to-fine object recognition network (CF-ORNet), as the second teacher network, which realizes fine-grained recognition through feature learning and category correction. After that, we propose a lightweight model from knowledge distillation to filter grafting on two teacher networks to achieve efficient fine-grained object recognition. The experimental results on Vehicle Detection in Aerial Imagery (VEDAI) and HR Ship Collection 2016 (HRSC2016) datasets achieve competitive performance.**

***Index Terms*— Coarse-to-fine object recognition network (CF-ORNet), filter grafting, fine-grained object recognition (FGOR), high-resolution remote sensing image (HR-RSI), knowledge distillation.**

I. Introduction

With the rapid development of high-resolution (HR) remote sensing satellite, imaging radar, and unmanned aerial vehicle (UAV) technology, the quantity and quality of remote sensing images (RSIs) have been dramatically enriched [1], [2], [3]. The fine-grained recognition of geospatial objects automatically distinguishes the categories or subcategories of geospatial objects in HR-RSIs, which has become a more practical and challenging task and has been widely utilized in many real-world applications [4], [5], [6], such as national defense security, disaster relief, nature protection, and urban construction.

Lately, deep learning-based techniques have dominated the highest accuracy benchmarks for various computer vision tasks, especially for fine-grained object recognition (FGOR) in natural images. Since the objects in HR-RSIs are generally observed from the overhead view, they often have the characteristics of various scales, arbitrary orientations, dense arrangement, and a few pixels covered [7]. Thus, it brings a great challenge to fine-grained object recognition in HR-RSIs. First, geospatial objects in HR-RSIs are arbitrary orientations and highly similar between fine-grained categories. Because only a limited number of pixels can be utilized to learn object features, it is not feasible to directly utilize the existing fine-grained object recognition methods. Hence, we need to include a rotation detector with oriented bounding boxes (OBBs) to adapt to arbitrary orientations. With the rapid development of deep learning methods, many object detection frameworks currently have excellent performance. For example, you only look once (YOLO) series [8], [9], [10] is prevailing in the object detection frameworks because of its fast detection speed and constantly updated network structure, but compared with other methods, there is still room for the accuracy of fine-grained objects recognition. Vision transformer (ViT) is very hot framework recently, but due to the lack of specific pretraining models for fine-grained recognition datasets as well as the low efficiency of mining discriminative features, it is inflexible to apply it to fine-grained objects recognition in HR-RSIs. In contrast, as one of the state-of-the-art (SOTA) rotation detectors, oriented region convolutional neural network (oriented R-CNN) [11] is more suitable for the tasks of fine-grained objects recognition, mainly due to the following advantages.

1) Oriented R-CNN is applied to rotation object detection as a two-stage method, which has higher detection accuracy compared with other methods.
2) Oriented R-CNN has overcome the computational bottleneck of generating a large number of rotation anchors in the region proposal network (RPN) with two-stage rotation detector.
3) The shared feature layer generated by oriented R-CNN in the detection head can be conveniently utilized for knowledge distillation (KL) and filter grafting.

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4) Choosing a detector with better performance can not only achieve higher location accuracy, but also reduce the difficulty of fine-grained category correction.

What is more, we expect to design an effective fine-grained learning strategy to improve the discrimination ability and representation granularity of object features. The available solutions are coarse-to-fine or repeated classification strategies.

In addition, the network structure of the deep model of fine-grained recognition is more complex, and its numerous parameters (Params) and dependence on computational power are not conductive to the deployment of practical engineering applications. That is, it is indispensable to bring into model lightweight technology [12], [13], [14] to reduce the computational cost of fine-grained object recognition. As we know, knowledge distillation [15], [16] has become a prevailing lightweight technology because of its excellent model compression ability. Unfortunately, limited by the ability of feature representation, the student network hardly breaks through the performance bottleneck. Therefore, how to strengthen the mutual learning of students in knowledge distillation to further fill the performance gap between students and teachers' network is a major challenge.

In summary, we propose an efficient fine-grained object recognition method in HR-RSIs from knowledge distillation to filter grafting. First, oriented R-CNN, a two-stage object detector is employed to achieve more accurate location and preliminary recognition of objects with arbitrary orientations. Then, we design a coarse-to-fine recognition strategy to improve the ability of fine-grained feature representation of objects. Furthermore, we propose a lightweight model from knowledge distillation to filter grafting to speed up the fine-grained object recognition in HR-RSIs. The main contributions of this study can be outlined as follows.

1) In view of various scales, dense arrangement, and highly similar objects in RSIs, a two-stage fine-grained object recognition network is designed to mine fine-grained features and category corrections of detector results to improve the recognition accuracy.
2) Due to the complex network structure and high computational cost of fine-grained object recognition, a lightweight model from knowledge distillation to filter grafting is proposed through channelwise knowledge distillation (CWD), weighted soft label knowledge distillation (WSLD), and filter grafting to reach efficient fine-grained object recognition in RSIs.
3) The proposed efficient model can better balance the precision and speed of two benchmark RSI datasets, including HR Ship Collection 2016 (HRSC2016) and Vehicle Detection in Aerial Imagery (VEDAI).

II. Related Work

A. Fine-Grained Object Recognition in RSIs

Fine-grained object recognition of HR-RSIs aims to distinguish the categories or subcategories of objects in RSIs, which can be divided into image-level recognition and object-level recognition. In fine-grained image recognition [17], [18], [19], object usually occupies a high proportion of pixels in the whole image. Thus, the focus is not on object location, but on mining discriminative features [20]. For example, Lin et al. [21] proposed a classic image-level fine-grained recognition algorithm, i.e., bilinear CNN structure. Zhang et al. [22] developed an attribute-guided multilevel enhanced feature representation network (AMEFRN) and an attribute learning network to learn the attribute information. In general, although the image-level recognition method achieves the purpose of mining discriminative features, it cannot perform well in fine-grained recognition of dense objects and lacks the reliability and stability of multiobject recognition because the location of geospatial objects is not considered.

In contrast, fine-grained object recognition needs to locate the interested objects in RSIs as accurately as possible to improve the discrimination and representation ability of object features, and then give a fine-grained classification of each object, simultaneously. There are many outstanding generic object detectors, such as faster R-CNN [23], single-shot multibox detector (SSD) [24], YOLO series [8], [9], [10], and RetinaNet [25]. In the remote sensing field, recent studies mostly train some prevalent deep learning-based architectures [26], [27], [28], [29] to build their models. For example, Li et al. [26] introduced polygon anchor to replace the traditional horizontal anchor and developed a rotation-insensitive RPN. In our previous work, a feature fusion and scaling-based single-shot detector (FS-SSD) for small object detection in UAV imagery was proposed [27] by adding feature pyramid network (FPN) variants to feature fusion single shot multi-box detector (FSSD) [30], improving the detection accuracy. Based on FS-SSD, we explored multiscale object detection in HR-RSI via rotation invariance deep features driven by oriented response FS-SSD and channel attention (OR-FS-SSD + CA) [7], which can learn object rotation as well as positively lift the performance of multiscale object detection. Since horizontal bounding boxes (HBBs) are inflexible for accurately locating geospatial objects with arbitrary orientations in RSIs, rotation detectors with OBBs have a promotion effect on the detection of arbitrary-oriented objects. For instance, Yang et al. [28] proposed refined rotation RetinaNet (R^3Det) imposing cascaded detection heads to improve the feature alignment ability of horizontal anchors for rotating objects. In our previous work, based on the RetinaNet, we proposed a structurally reparameterized rotation detector (SRep-RDet) [29] for arbitrary-oriented objects in HR-RSIs by introducing RepVGG-B1g2 and channel attention mechanism, achieving competitive performance on dataset for object detection in aerial images (DOTA) and HRSC2016 datasets. In addition to the abovementioned one-stage detector, the two-stage method has dominance in detection precision. Xie et al. [11] brought into oriented R-CNN to directly convert from horizontal anchor boxes to predicted oriented region of interests (RoIs), which reduces the redundancy of the two-stage detector and has balance in precision and speed. As a result, we think oriented R-CNN is more suitable for accurately locating objects in HR-RSIs.

In fine-grained object recognition tasks, Liu et al. [31] proposed a rotation detector based on fast R-CNN to detect ship objects and then utilized the location and classification

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branches of the detector to achieve fine-grained recognition of ship objects. To further extend the recognizable objects in RSIs, Ming et al. [32] proposed oriented feature alignment network (OFA-Net) through a rotated bounding boxes refinement module to achieve better fine-grained oriented object recognition, which only extracts local feature; there is still room for improvement in feature granularity. Relatively, Sun et al. [6] introduced a cascaded hierarchical object detection network (CHODNet) for fine-grained object recognition in RSIs relying on oriented location and coarse-to-fine classification, which can be viewed as a baseline model on a novel benchmark dataset. The generalization ability of the model can be further improved by mining multilevel granularity features.

From the abovementioned studies, we can see that fine-grained object recognition methods require mining more fine-grained content for each object. A feasible idea is to combine object location and fine-grained classification by imposing the strategy of coarse-to-fine or repeated classification, so as to improve the discrimination ability and representation granularity of object features. Consequently, we will design a fine-grained object recognition network in HR-RSIs based on a coarse-to-fine recognition strategy to fully extract the finer grained features of each object and correct the misclassification in the preliminary recognition.

B. Lightweight Network

Because of its complex structure, many Params, and dependence on computational power, deep model is inflexible to be directly applied to practical engineering. Some lightweight techniques, such as model compression and mutual learning, have been proposed to drop the deep model Params while placing them slightly within the proper performance range.

Model compression techniques mainly include pruning, quantization, and knowledge distillation [33]. Among them, knowledge distillation is a training strategy to transfer knowledge from a high performance, large teacher network to a smaller student network. For instance, Wang et al. [34] designed a network that simultaneously trains teacher and student to narrow the gap between student and teacher networks by comparing the differences in the feature maps around the anchor box. Shu et al. [15] proposed a channelwise distillation (CD) method for object detection or semantic segmentation, which imparts the teacher's knowledge to the student network to achieve knowledge distillation. Yue et al. [35] introduced an adaptive distillation method based on 3-D transformer, using self-supervised method to train the network to achieve the classification of hyperspectral images. In [36], to improve the recognition efficiency of porn streamers in live video, we developed a deep model compression method based on multimodal knowledge distillation, which can effectively reduce the computational cost and improve the recognition speed.

Another lightweight technique is to change the training way to promote the network performance, in which mutual learning is a vital supplement to the lightweight network with limited performance. Zhang et al. [37] proposed a model training strategy of mutual learning by simultaneously training two or more identical models and sharing the information between these models to achieve a performance improvement. Meng et al. [38] introduced a filter grafting technique to reactivate the invalid weights in the model and utilized a weight-based entropy criterion to realize the grafting of multiple models. Although the mutual learning method works very well in models with limited performance, this improvement is more partial; there is a problem of weak generalization ability.

In general, knowledge distillation as an effective lightweight technique can be deployed on deep model and accelerated by compressing the model Params. Conversely, mutual learning and filter grafting as a multinetwork parallel training strategy can further improve the network performance. Considering that filter grafting is a model interactive training method, it will further fill the performance gap between student and teacher networks if combined with filter grafting. We propose an efficient fine-grained object recognition model from knowledge distillation to filter grafting, which aims to speed up the inference process while ensuring precision.

III. Method

A. Overall

Our overall structure is composed of two parts: fine-grained object recognition and lightweight model, as shown in Fig. 1. In part A, we design a two-stage fine-grained object recognition network, including rotation object detection and coarse-to-fine object recognition network (CF-ORNet). In part B, we combine knowledge distillation and filter grafting to lightweight fine-grained object recognition model.

Part A is the two-stage fine-grained object recognition, in which Stage 1 is to obtain the accurate location of arbitrary-oriented objects and preliminary classification results by employing the oriented R-CNN detector, and Stage 2 is to build the CF-ORNet to realize fine-grained object recognition by fully mining the features of each object and correcting the error category in the preliminary recognition.

Part B is to lightweight the networks of Part A from knowledge distillation to filter grafting, in which the two teacher networks of Part A are applied to guide efficient student networks through knowledge distillation, and then, filter grafting in knowledge distillation is conducted to complete the mutual learning between student networks.

B. Fine-Grained Object Recognition

For fine-grained object recognition, we design a two-stage coarse-to-fine object recognition strategy. First, Stage 1 employs the oriented R-CNN [11] detector for object location and preliminary classification. The main utility of oriented R-CNN is to achieve accurate location as a baseline rotation detector and to guide the compact process of the network as a teacher network. Then, Stage 2 constructs a CF-ORNet by imposing a coarse-to-fine strategy for fine-grained feature representation and category correction.

The working process of Stage 1 oriented R-CNN teacher network can be depicted as follows.

Step 1: Multiscale features $\mathbf{P}\{P_2, P_3, P_4, P_5, P_6\}$ are extracted from Block 1–Block 4 by residual network-101 (ResNet-101) backbone-FPN and fused into feature $\mathbf{M} \in \mathbb{R}^{H\times W\times 256}$.

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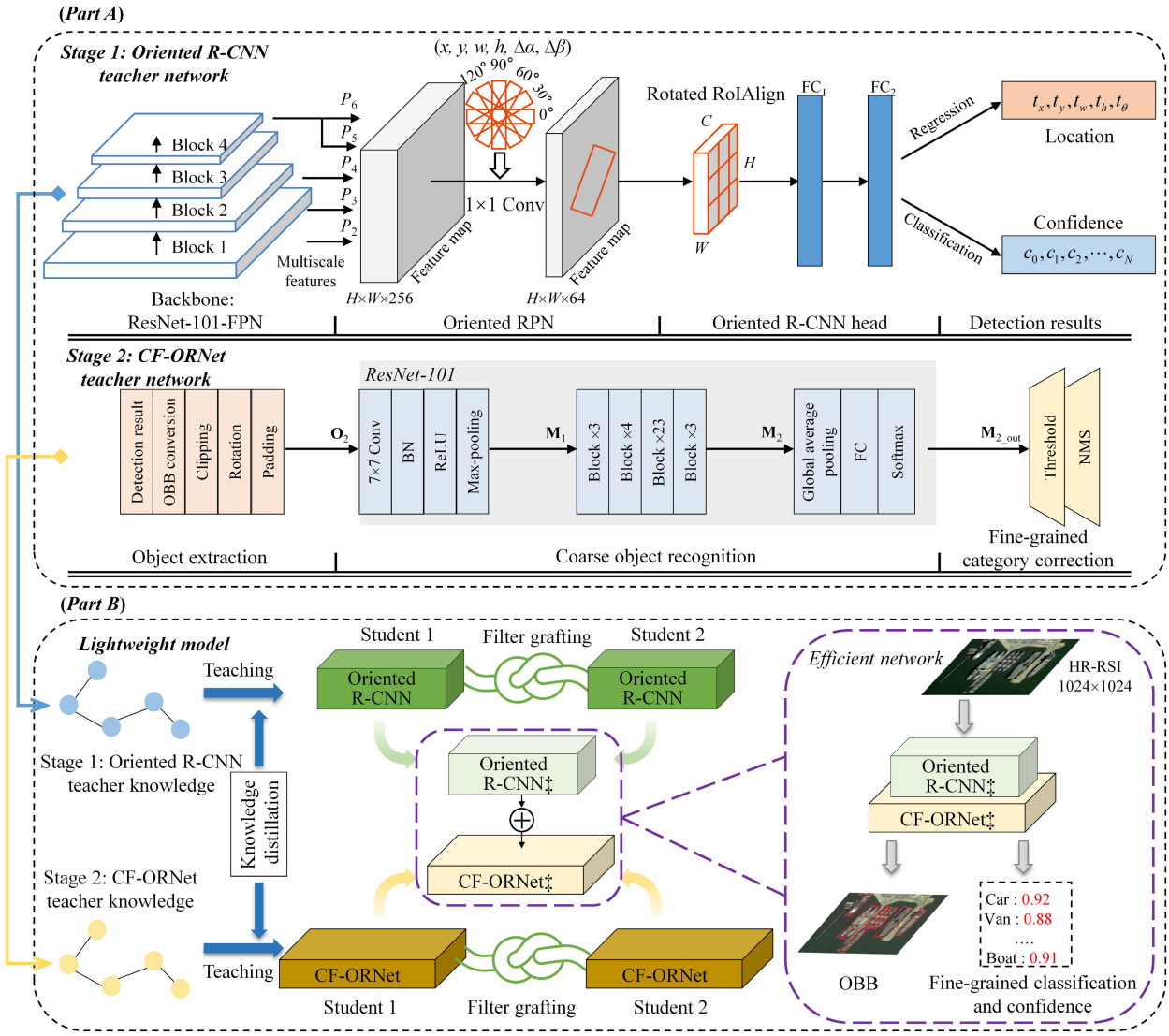


Fig. 1. Overall structure of the proposed fine-grained object recognition network in RSIs. Part A is two-stage fine-grained object recognition, in which Stage 1 is to obtain the accurate location of arbitrary-oriented objects and preliminary classification results by employing the oriented R-CNN detector as the teacher network, and Stage 2 is to build the CF-ORNet teacher network to realize fine-grained object recognition. Part B is to lightweight the networks of Part A from knowledge distillation to filter grafting.

Step 2: After compressing $\mathbf{M} \in \mathbb{R}^{H\times W\times 256}$ into $\mathbb{R}^{H\times W\times 64}$ by 1×1 Conv, oriented RPN generates a series of high-quality oriented proposals (x, y, w, h, $\Delta\alpha$, $\Delta\beta$) in a nearly cost-free manner.

Step 3: By utilizing the multiscale features $\mathbf{P}\{P_2, P_3, P_4, P_5, P_6\}$ and oriented proposals (x, y, w, h, $\Delta\alpha$, $\Delta\beta$), oriented R-CNN head is designed to refine the RoIs and identify them through rotated RoIAlign and two fully connected layers (FC_1 and FC_2) (see Fig. 1).

Step 4: Through regression and classification, the detection results mainly include two parts: 1) the accurate location (t_x, t_y, t_w, t_h, t_θ) of geospatial object will be utilized to extract objects in RSIs and 2) the initial category and confidence (c_0, c_1, c_2, ..., c_N) of geospatial object will provide an empirical reference for the classification in the subsequent modules.

In Stage 2, based on the location and preliminary classification results of rotation detector, our CF-ORNet can realize fine-grained object recognition by imposing a coarse-to-fine recognition strategy, which is divided into three modules: object extraction, coarse object recognition, and fine-grained category correction. Fig. 2 illustrates the inference process.

1) Object Extraction: The object extraction of CF-ORNet includes OBB conversion, clipping, rotation, and padding, in which the location and the initial category can be obtained through the rotation detector.

Step 1: By analyzing the preliminary detection results that are composed of location and initial category G_i, the location results are converted into OBBs as

$$\text{OBB} = (x, y, w, h, \theta)$$
$$\text{where} \begin{cases} x = \dfrac{x_1 + x_3}{2}, \quad y = \dfrac{y_1 + y_3}{2} \\ w = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ h = \sqrt{(x_4 - x_3)^2 + (y_4 - y_3)^2} \\ \theta = \arctan\left(\dfrac{y_2 - y_1}{x_2 - x_1}\right) \end{cases} \tag{1}$$

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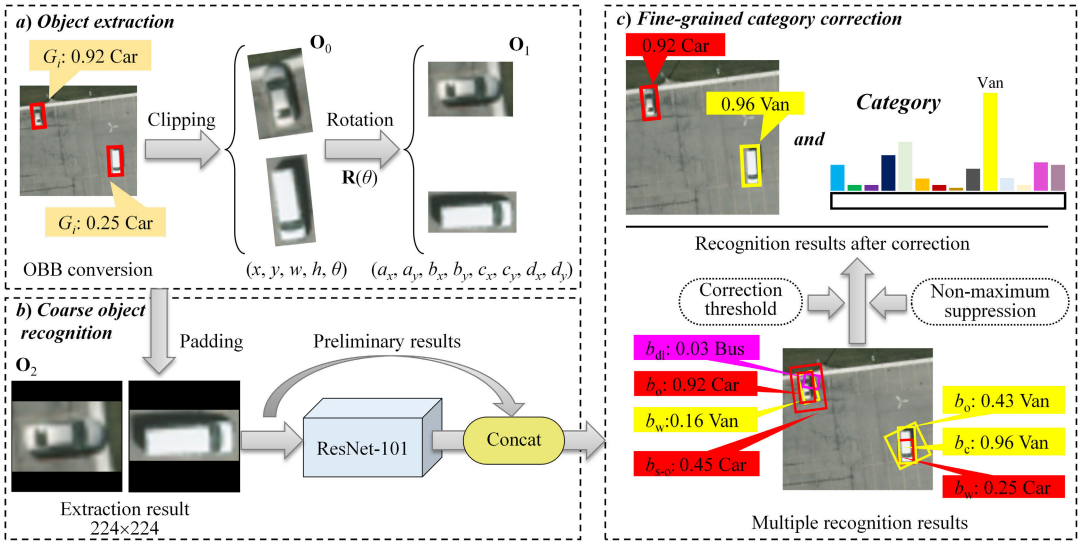


Fig. 2. Inference process of CF-ORNet: (a) object extraction of CF-ORNet includes OBB conversion, clipping, rotation, and padding, where the location and the initial category can be obtained through rotation detector; (b) coarse object recognition. ResNet-101 structure is introduced to mine stronger discriminative features of fine-grained classification; and (c) fine-grained category correction. For the fine-grained recognition results mixed with the preliminary recognition results, the module is designed to correct the category by setting the category correction threshold.

where x and y are the horizontal and vertical coordinates of the OBB center point, w is the length of the long side, h is the length of the short side, θ is the angle between the side and the horizontal positive half-axis, and the value range is $[-\pi/4, -3\pi/4]$.

Step 2: Clip the located objects into an object-patch $\mathbf{O}_0$ while recording the categories and confidence of each object. The confidence here reflects the degree of certainty that the object belongs to a certain category, and the quality of the objects' location.

Step 3: Considering that OBB with arbitrary orientations (x, y, w, h, θ) is tough for fine-grained classification, it is rotated to the horizontal direction $(a_x, a_y, b_x, b_y, c_x, c_y, d_x, d_y)$ to effectively weaken the impact as well as reduce the learning difficulty of classification network. The horizontal object-patch $\mathbf{O}_1$ can be obtained by calculating the rotation matrix $\mathbf{R}(\theta)$ for each OBB

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & \dfrac{w\cos\theta + h\sin\theta - w}{2} \\ -\sin\theta & \cos\theta & \dfrac{w\sin\theta + h\cos\theta - h}{2} \end{bmatrix}. \quad (2)$$

Step 4: Since the input size of the classification network is usually set to 224px × 224px, the long and short sides are scaled and filled to 224px by padding without distortion to generate object-patch $\mathbf{O}_2$.

2) Coarse Object Recognition: Next, we put forward utilizing ResNet-101 structure to mine stronger discriminative features for fine-grained classification. The inference process is as follows.

Step 1: Object-patch $\mathbf{O}_2$ carrying geospatial objects passes 7 × 7 Conv, batch normalization (BN), rectified linear unit (ReLU), and max-pooling for fast compression.

Step 2: Compressed feature $\mathbf{M}_1$ inputs the blocks of $\{\times 3, \times 4, \times 23, \times 3\}$ to extract the deep feature $\mathbf{M}_2$.

Step 3: Through global average pooling, FC, and softmax, $\mathbf{M}_{2_out} \in \mathbb{R}^{H\times W\times D}$ is regarded as the final feature map of object recognition containing important information of fine-grained classification.

3) Fine-Grained Category Correction: For unknown categories, they are mixed with the preliminary recognition results to correct the categories. Fig. 3 illustrates the preliminary recognition results of the rotation detector before category correction. Among them, three objects are recognized as Nimitz-class aircraft carriers, while only the object with a confidence of 0.9781 produces the correct recognition result, and the other two objects with lower confidence have incorrectly located and classified, respectively.

Therefore, it is necessary to set the category correction threshold for error recognition. We only pick the preliminary object recognition results with confidence less than threshold for category correction. The correction process is shown in Fig. 2 and described as follows.

Step 1: After mixing the preliminary recognition result and secondary recognition result, the existing bounding boxes will include: 1) the optimal boxes b_o and the suboptimal boxes b_{s-o} from rotation detector; 2) the boxes with deviated location b_{dl} generated in the process of oriented location; and 3) the optimal correct boxes b_c newly generated after category correction and the optimal wrong boxes b_w retained from the preliminary recognition result, which will provide location, category, and confidence information for the correction process.

Step 2: The correction threshold retains the optimal correct boxes and wrong boxes of b_o, b_c, and b_w as the recognition result, and then filters the fine-grained object recognition results by using nonmaximum suppression (NMS) to remove redundant boxes. Here, we use the oriented intersection over union (IoU) to obtain the overlap between the highest confidence bounding box and other bounding boxes

$$\mathrm{IoU}(A, B) = \frac{S(A) \cap S(B)}{S(A) \cup S(B)} \quad (3)$$

where S is the area contained in the OBB.

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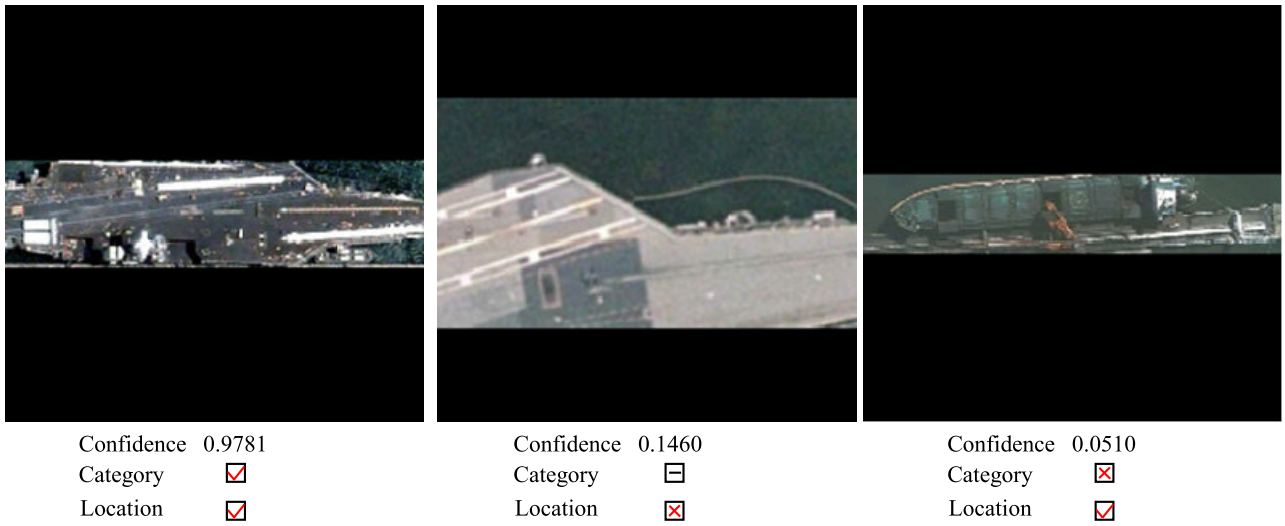


Fig. 3. Preliminary recognition results of the detector. Three objects are recognized as Nimitz-class aircraft carriers, while only object with a confidence of 0.9781 produces correct recognition result, and the other two objects with a lower confidence are incorrectly located and classified, respectively.

So far, we have retained the correct result of the coarse classification information of the detector, and the error categories are corrected to the fine-grained categories with higher confidence.

4) Loss Function Design: The fine-grained object recognition network in RSIs needs accurate location and fine-grained classification. Based on our coarse-to-fine classification strategy, our loss function includes three parts: object location loss, object coarse-classification loss, and object fine-classification loss. For the object location loss, we propose using the smooth-L_1 loss to penalize the misalignment between the predicted bounding boxes and the ground-truth ones

$$F_{\text{smooth-}L_1}(x) = \begin{cases} 0.5^2 x, & |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{4}$$

where x is the absolute difference between the predicted and true values, and the object location loss is

$$\mathcal{L}_{\text{loc}} = \frac{1}{N_{\text{pos}}} p_i^* \sum_{i=1}^{N} F_{\text{smooth-}L_1}\left(X_i, G_i^*\right) \tag{5}$$

where N_{pos} is the number of positive samples, p_i^* is the ground-truth label of the ith anchor, N is the total number of samples, and X_i and G_i^* are the offsets of the anchor corresponding to the ith sample and ground-truth box, respectively.

For the loss of coarse-classification and fine-classification of object, we use the cross-entropy loss, calculated as

$$F_{\text{cls}} = \frac{1}{N_{\text{pos}}} \sum_{N_{\text{pos}}} \sum_{i} -y_i \log P_i \tag{6}$$

where y_i is the label of the ground-truth box, P_i is the classification confidence of category i, and the coarse-classification loss is acquired as

$$\mathcal{L}_{\text{coarse}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N} F_{\text{cls}}\left(c_i^{\text{C}}, l_i\right) \tag{7}$$

where c_i^{C} is the output of the classification branch of oriented RPN, which denotes the probability that the proposal belongs to the object, l_i is the ground-truth label of the ith anchor, and the fine-classification loss is generated as

$$\mathcal{L}_{\text{fine}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N} F_{\text{cls}}\left(c_i^{\text{F}}, R_i\right) \tag{8}$$

where c_i^{F} and R_i denotes the classification confidence and ground-truth label of the ith sample of the classifier, respectively.

CF-ORNet utilizes the fine-classification loss function as the classification penalty for the geospatial objects extracted from the image, while the training for oriented R-CNN is based on the source dataset. Since oriented R-CNN has made a preliminary decision, it is necessary to compensate and correct the results by using the proposed CF-ORNet by imposing a coarse-to-fine strategy. Finally, through the comprehensive judgment of classification results, more accurate fine-grained object recognition results can be obtained

$$\text{FGOR} = (1 - \lambda)\mathcal{P}_{\text{coarse}} + \lambda \mathcal{P}_{\text{fine}} \tag{9}$$

where λ is the correction threshold and $\mathcal{P}$ is the classification result.

C. Lightweight Model From Knowledge Distillation to Filter Grafting

The performance of deep networks generally benefits from numerous Params, but due to their complex structure and dependence on computing cost, there is a "balance" issue between precision and speed in large networks. In this article, we are committed to balancing the recognition precision and speed of fine-grained object recognition, and improving the recognition efficiency while ensuring proper precision. As shown in Fig. 4, in view of both location precision and classification accuracy that consuming a lot of computing cost, we combine object detection and fine-grained classification knowledge through the distillation method of CWD and WSLD to train the model with fewer Params and faster recognition speed. However, this lightweight method will

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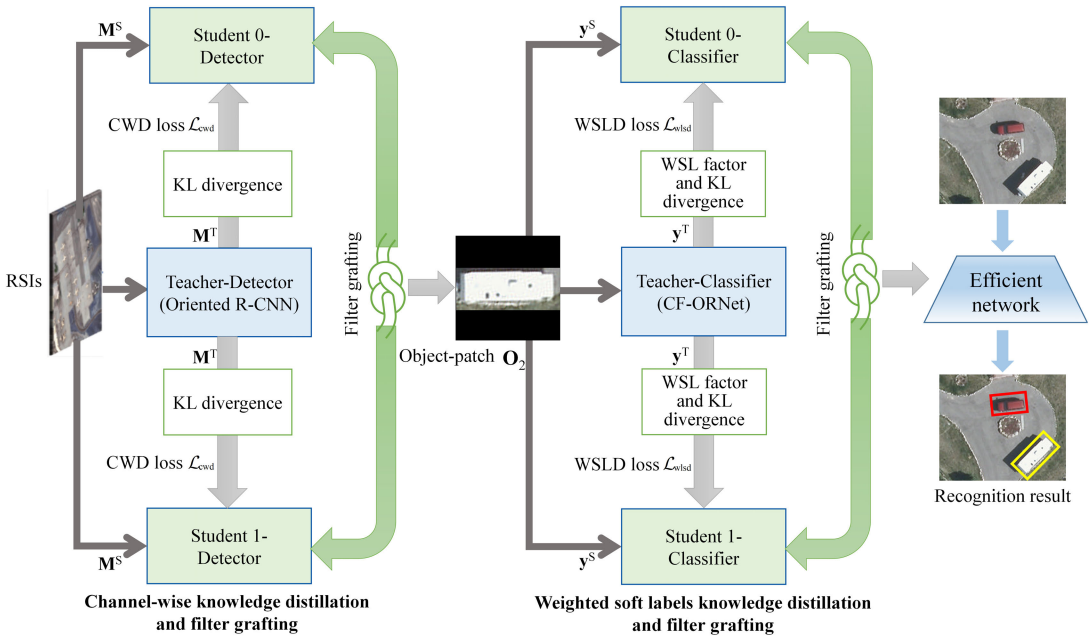


Fig. 4. Lightweight model from knowledge distillation to filter grafting. CWD and WSLD are utilized to train the student networks of rotation detector and CF-ORNet, respectively. Afterward, filter grafting is deployed in knowledge distillation to further fill the performance gap between student and teacher networks.

inevitably reduce the performance of compact networks. To fill the performance gap, we propose to introduce filter grafting method deployed in the training process of student networks.

1) Channelwise Knowledge Distillation: CWD is specially designed for oriented R-CNN detector, and its implementation is as follows.

Step 1: After sending the RSI into the teacher and student detector simultaneously, corresponding feature maps $\mathbf{M}^{\mathrm{T}}$ and $\mathbf{M}^{\mathrm{S}}$ are generated, respectively.

Step 2: Referring to the design form of softmax, the following equation denotes the probability distribution of feature map in different channels to calculate KL divergence. Specifically, by applying softmax normalization, we eliminate the influences of magnitude scales between large networks and compact networks. Since this normalization facilitates knowledge distillation, the probability distribution $\Phi(\mathbf{M})$ is calculated as

$$\Phi(\mathbf{M}) = \frac{\exp\left(\frac{\mathbf{M}_{c,i}}{\tau}\right)}{\sum_{i=1}^{W\cdot H}\exp\left(\frac{\mathbf{M}_{c,i}}{\tau}\right)} \tag{10}$$

where $\mathbf{M}_{c,i}$ is the value of the output feature map at the ith spatial position of the cth channel, $c \in [1, C]$, C is the total number of feature map channels; $i \in [1, W \cdot H]$, W and H are the width and height of the feature map, respectively; and τ is a hyper-parameter (the temperature). By converting $\mathbf{M}$ into $\Phi(\mathbf{M})$, it is more convenient to measure the KL divergence distance φ between $\mathbf{M}^{\mathrm{T}}$ and $\mathbf{M}^{\mathrm{S}}$

$$\varphi\left(\mathbf{M}^{\mathrm{T}}, \mathbf{M}^{\mathrm{S}}\right) = \frac{\tau^2}{\mathbf{C}}\sum_{c=1}^{C}\sum_{i=1}^{W\cdot H}\Phi\left(\mathbf{M}_{c,i}^{\mathrm{T}}\right)\cdot\log\left[\frac{\Phi\left(\mathbf{M}_{c,i}^{\mathrm{T}}\right)}{\Phi\left(\mathbf{M}_{c,i}^{\mathrm{S}}\right)}\right]. \tag{11}$$

The goal of distillation is to minimize $\varphi(\mathbf{M}^{\mathrm{T}}, \mathbf{M}^{\mathrm{S}})$. Therefore, when the teacher produces a probability distribution value $\Phi(\mathbf{M}_{c,i}^{\mathrm{T}})$, the probability distribution value $\Phi(\mathbf{M}_{c,i}^{\mathrm{S}})$ of the student network at the same position should approach $\Phi(\mathbf{M}_{c,i}^{\mathrm{T}})$ to realize the guidance of teachers to students.

Step 3: The teacher–student distance $\varphi(\mathbf{M}^{\mathrm{T}}, \mathbf{M}^{\mathrm{S}})$ can be utilized as the knowledge distillation penalty item in the loss function $\mathcal{L}_{\mathrm{cwd}}$, and form the overall loss function with the student loss function $\mathcal{L}_{\mathrm{self}}\left(\mathbf{G}, \mathbf{M}^{\mathrm{S}}\right)$. The overall loss function is as follows:

$$\mathcal{L}_{\mathrm{cwd}} = \mathcal{L}_{\mathrm{self}}\left(\mathbf{G}, \mathbf{M}^{\mathrm{S}}\right) + \alpha\cdot\varphi\left(\mathbf{M}^{\mathrm{T}}, \mathbf{M}^{\mathrm{S}}\right) \tag{12}$$

where α is the hyper-parameter balancing knowledge distillation loss and self-learning loss.

In short, the lightweight detector gains accelerated inference capability through CWD. At the same time, the object recognition performance of lightweight model is further improved under the guidance of teachers.

2) Weighted Soft Label Knowledge Distillation: WSLD is specifically designed for CF-ORNet, and the detailed operation steps are as follows.

Step 1: For the same object-patch $\mathbf{O}_2$, the teacher and student classifiers will obtain two classification vectors $\mathbf{y}^{\mathrm{T}}$ and $\mathbf{y}^{\mathrm{S}}$, respectively, to distinguish the image categories.

Step 2: Weighted soft label (WSL) loss is the product of two multiplier terms, where the first term is the WSL factor that dynamically adjusts the proportion according to the predictions of teacher and the student, and the second term is the general knowledge distillation loss L_{kd} that uses the classification results of teacher to guide the training of student. The WSL loss is defined as

$$\mathcal{L}_{\mathrm{wsl}} = \left[1 - \exp\left(-\frac{\mathcal{L}_{\mathrm{CE}}^{\mathrm{S}}}{\mathcal{L}_{\mathrm{CE}}^{\mathrm{T}}}\right)\right]\left(-\tau^2\sum_{k} y_k^{\mathrm{T}}\log y_k^{\mathrm{S}}\right) \tag{13}$$

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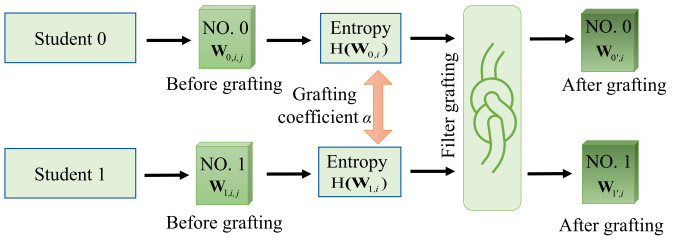


Fig. 5. Details of filter grafting between student networks. Two student networks (Student 0 and Student 1) are trained in parallel. The information entropy of the convolution layer is calculated for the filter grafting of the two networks.

where y_k^{T} and y_k^{S} represent the kth element of the classification vector output by teacher and student, τ is a hyper-parameter, and $\mathcal{L}_{\mathrm{CE}}^{\mathrm{S}}$ and $\mathcal{L}_{\mathrm{CE}}^{\mathrm{T}}$ are the cross-entropy loss functions of student and teacher, respectively.

Step 3: The knowledge distillation loss and the student's self-learning loss are combined to iteratively optimize in the training process. The final loss function is

$$\mathcal{L}_{\mathrm{wsld}} = \mathcal{L}_{\mathrm{CE}}^{\mathrm{S}} + \alpha \mathcal{L}_{\mathrm{wsl}} \tag{14}$$

where α is the hyper-parameter balancing knowledge distillation loss and self-learning loss.

The knowledge of the teacher network is transferred to ResNet-18 or other effective networks through WSLD, so that the student network can better mine the discriminative feature information of objects.

3) Filter Grafting: Filter grafting is a multinetwork parallel training strategy for compressing deep neural network, which can reduce the parameter and computational complexity without sacrificing the model accuracy. However, compared with teacher (large) networks, student (compressed) networks using knowledge distillation have limitations in feature representation. To bridge the gap between the teacher network and the student network, we introduce a filter grafting between the two student networks of the detector and the classifier to maintain the original accuracy as much as possible, as shown in Fig. 5.

The details are as follows.

Step 1: Two student networks (Student 0 and Student 1) are trained in parallel, which can be two student detectors or two student classifiers. After each training epoch, the filter will record the weights $\mathbf{W}_{N,i,j}$ of each convolution layer of Student 0 and Student 1, which represent the weight value of the jth neuron in the ith convolution layer of Student N ($N \in [0, 1]$).

Step 2: The difference between the maximum and minimum values of all neurons in the ith convolution layer is calculated and divided into ten intervals to produce the discrete probability distribution $p_i = \{p_1, \ldots, p_K\}$, where $K \in [1, 10]$. Since entropy reflects the amount of information contained in the convolution layer, the information entropy $\mathbf{H}(\mathbf{W}_{N,i})$ of the convolution layer is calculated through the probability distribution p_i

$$\mathrm{H}\left(\mathbf{W}_{N,i}\right) = -\sum_{i=1}^{10} p_i \log p_i. \tag{15}$$

Step 3: After obtaining the entropy of the convolution layer, the two networks can be grafted. We choose the layer $\mathbf{W}_{N',i}$ as the basic unit of grafting to maintain the continuity of the weights in the layer. Grafting in layers is as follows:

$$\mathbf{W}_{N',i} = \alpha \mathbf{W}_{N,i} + (1-\alpha)\mathbf{W}_{N\%1,i} \tag{16}$$

where % is the modular operation, α is the grafting coefficient, and $\mathbf{W}_{N,i}$ and $\mathbf{W}_{N',i}(N \in [0,1])$ represent the weights of convolution layers before and after filter grafting, respectively. Notably, grafting is a mutual process, which can conduct from Student 0 to Student 1 or from the other direction. Grafting coefficient α is a key parameter to determine the grafting direction

$$\alpha = A \cdot \arctan\left(c\left(\mathrm{H}\left(\mathbf{W}_{N,i}\right) - \mathrm{H}\left(\mathbf{W}_{N\%1,i}\right)\right)\right) + 0.5 \tag{17}$$

where A and c are the hyper-Params.

Through the parallel training of filter grafting, the student networks have effectively completed mutual learning, that is, the weights in the convolutional layer of different student networks are complementary to obtain richer feature information, and thus, the student networks of detector and CF-ORNet can achieve higher accuracy.

IV. Experimental Results and Analysis

A. Experimental Setup

1) Dataset: To verify the effectiveness, we conduct extensive experiments on HRSC2016 dataset [31] and VEDAI dataset [39] that are most widely used on fine-grained object recognition in RSIs. The characteristics of two datasets are described in detail in the following.

HRSC2016 dataset contains 1061 images with the size ranging from 300 × 300 to 1500 × 900, in which a total of 2976 ship objects. We recognize 22 categories of fine-grained ship objects and adopt the HRSC2016 official dataset division scheme to obtain 436 images for training, 181 images for validation, and 438 images for testing.

VEDAI dataset contains 1210 RSIs with the size of 1024 × 1024 and a total of 3640 vehicle objects with oriented boxes annotation. We recognize six types of fine-grained vehicle objects, and the VEDAI dataset is divided into training set and test set according to the ratio of 4:1, i.e., 968 images for training and 242 images for testing.

2) Implementation Details:

a) Experiment settings: The experiments are conducted on Ubuntu16.04 operating system, an NVIDIA GeForce RTX 3080 GPU, and the PyTorch deep learning framework. We optimize the overall network with the stochastic gradient descent (SGD) algorithm with the initial learning rate of 0.005 and adopt the learning rate warm-up in the first 500 iterations. The momentum and weight decay are set to 0.9 and 0.0001, respectively. The rotation detector employs oriented R-CNN structure, and ResNet-101, ResNet-50, and ResNet-18 are used as our backbone. The backbone of all methods is pretrained on ImageNet [40]. In knowledge distillation, we set the maximum number of students in the filter grafting to 2, the hyper-parameter A in (17) to 0.4, and the hyper-parameter c to 500. In the training, the oriented R-CNN is utilized for

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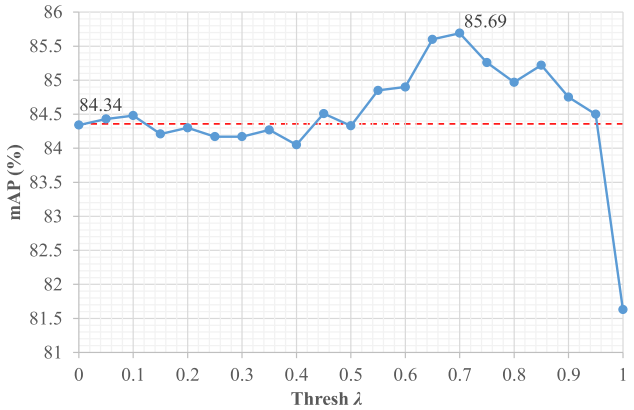


Fig. 6. Effect of different threshold λ on performance. When the threshold is 0.7, it reaches the highest mAP (85.69%), 1.35% higher than the initial recognition result.

accurate location and preliminary classification of the objects in Stage 1, which will first trained as a rotation detector and a teacher network. Unlike oriented R-CNN, CF-ORNet is trained in Stage 2 as the key to the coarse-to-fine recognition strategy and another teacher network. In the inference process, oriented R-CNN and CF-ORNet jointly utilize the inference process.

b) Evaluation indicators: The mean average precision (mAP) and frames per second (FPS) are utilized to evaluate object recognition performance, and the Params and floating-point operations per second (FLOPs) are used to evaluate the model complexity.

B. Experiment I: Effect of Category Correction Threshold

We compare the performance (mAP) of CF-ORNet with different category correction thresholds λ on the HRSC2016 dataset to explore the best threshold. As shown in Fig. 6, to ensure the fairness, we use the preliminary object recognition results from oriented R-CNN detector as the baseline mAP (84.34%, red line in Fig. 6), i.e., the threshold is set 0. When the threshold is less than or equal to 0.5, the mAP fluctuates near the red line with the increase of threshold. If the selected threshold is greater than 0.5, the preliminary recognition result is greatly improved. If the threshold is 0.7, it reaches the highest mAP (85.69%), which is 1.35% higher than the initial recognition result. However, the mAP drops sharply when the threshold is greater than 0.95.

The results show that setting the threshold within a reasonable range can evidently improve the recognition accuracy, but beyond the threshold range, the model accuracy will decline rapidly because of the dependence of CF-ORNet. After analysis, the two tasks of location and classification in fine-grained object recognition are complementary.

C. Experiment II: Comparison of CF-ORNet With Different Backbone

We compare the mAP of CF-ORNet with different backbones in object recognition performance on HRSC2016 dataset. In the experiment, we use five SOTA networks (ResNeSt-50 [41], Swin-Transformer [42], DeiT [43], ConvNeXt [44], and ResNet-101 [45]) as the backbone of CF-ORNet and take the result of oriented R-CNN detector as the baseline. The recognition results with different backbone are illustrated in Table I, in which CF-ORNet with ResNet-101 gains the best performance, achieving the highest mAP (85.69%) on the HRSC2016 dataset, 1.35% higher than the baseline. As well as, CF-ORNet with ConvNeXt and DeiT has a good improvement effect, which is 1.18% and 0.72% of mAP, respectively, higher than the baseline. ResNeSt-50 and Swin-Transformer also present some improvements. Comparing with the baseline, mAP increases by 0.29% and 0.35%, respectively.

TABLE I
RECOGNITION PERFORMANCE OF DIFFERENT BACKBONE

Backbone	mAP(%)	Upgrade
None	84.34	—
ResNeSt-50 [41]	84.63	0.29
Swin-Transformer [42]	84.69	0.35
DeiT [43]	85.06	0.72
ConvNeXt [44]	85.52	1.18
ResNet-101 [45]	**85.69**	**1.35**

Through the above results, although ResNet-101 is a classic classification network, it still shows powerful performance in fine-grained feature extraction of geospatial objects. Table I verifies that our CF-ORNet can improve the performance of preliminary object recognition no matter what backbone is utilized, which shows that our coarse-to-fine strategy is effective to mine the discriminative features of objects.

D. Experiment III: Visual Results on HRSC2016 and VEDAI Datasets

We visualize a subset of the experimental results and generate recognition results. As displayed in Figs. 7 and 8, we, respectively, show the recognition results of our method on different backbones (ConvNeXt, DeiT, and ResNet-101) on HRSC2016 and VEDAI datasets. For the OBB results based on oriented R-CNN, there are partially overlapping bounding boxes in both datasets, such as the first and third images in Fig. 7. This is because of the high similarity between objects in fine-grained object recognition, and the oriented R-CNN detector cannot sufficiently extract the discriminative features. In the second image in Fig. 7, the detection results of OBB and CF-ORNet with ConvNeXt and DeiT are misclassified at the dock. In the results of CF-ORNet with ResNet-101, the dock area is correctly classified by category correction. For the Austin warship in the third image in Fig. 7, the bounding boxes overlap appears in the OBB result, while the object is correctly classified after using DeiT and ResNet-101 in CF-ORNet. For the first image in Fig. 8, the truck object on the right side of the OBB result is not correctly classified, but CF-ORNet with three different backbones can correctly recognize truck by using category correction. For the second image in Fig. 8, in the OBB results and the recognition results of CF-ORNet with ConvNeXt and DeiT, a building in the middle area of the image is incorrectly recognized as a camper, and

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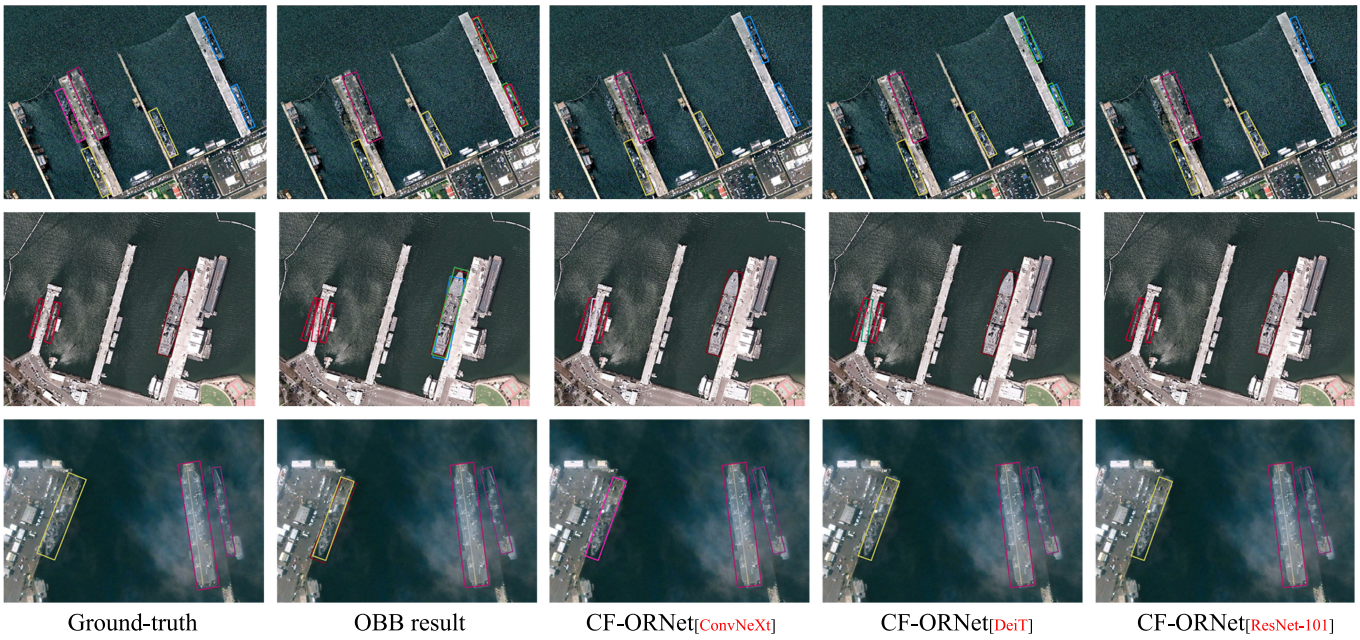


Fig. 7. Recognition results of different backbones on the HRSC2016 dataset. (From the first column to the last column) Ground truth, OBB result, CF-ORNet with ConvNeXt backbone, CF-ORNet with DeiT backbone, and CF-ORNet with ResNet-101 backbone.



Fig. 8. Recognition results of different backbones on the VEDAI dataset. (From the first column to the last column) Ground-truth, OBB result, CF-ORNet with ConvNeXt backbone, CF-ORNet with DeiT backbone, and CF-ORNet with ResNet-101 backbone.

only CF-ORNet with ResNet-101 successfully eliminates the region. In a word, choosing a suitable backbone is very critical for the final fine-grained recognition results of CF-ORNet.

E. Experiment IV: Comparison With SOTA Methods

In this experiment, we compare the object recognition performance of our method with the baseline, including shape-adaptive selection and measurement (SASM) [46], SRep-RDet [29], single-shot alignment network (S^2A-Net) [47], rotation-equivariant detector (ReDet) [48], and oriented R-CNN [11], on HRSC2016 and VEDAI datasets. As depicted in Fig. 9 and Table II, to prove the effectiveness of CF-ORNet, we explore the performance of the above methods combined with CF-ORNet, respectively. Oriented R-CNN with ResNet-101 obtains the highest mAP (84.34%) on HRSC2016, while ReDet gains the highest mAP (77.55%) but the lowest FPS (5.4) on VEDAI dataset. After imposing our coarse-to-fine recognition strategy, our CF-ORNet almost improves the fine-grained recognition performance of all detectors on the two datasets except ReDet on the VEDAI dataset because of the lack of structural consistency between the backbone (ReResNet) of ReDet and backbone (ResNet) of CF-ORNet. The oriented R-CNN + CF-ORNet method achieves the highest mAP (85.69% and 77.34%) on both datasets. In particular, the mAP of SASM promotes 7.32% after introducing CF-ORNet. The high improvement of model performance demonstrates the amazed potential of our CF-ORNet. In terms of speed, oriented R-CNN + CF-ORNet reaches a relatively fast FPS (13.7),

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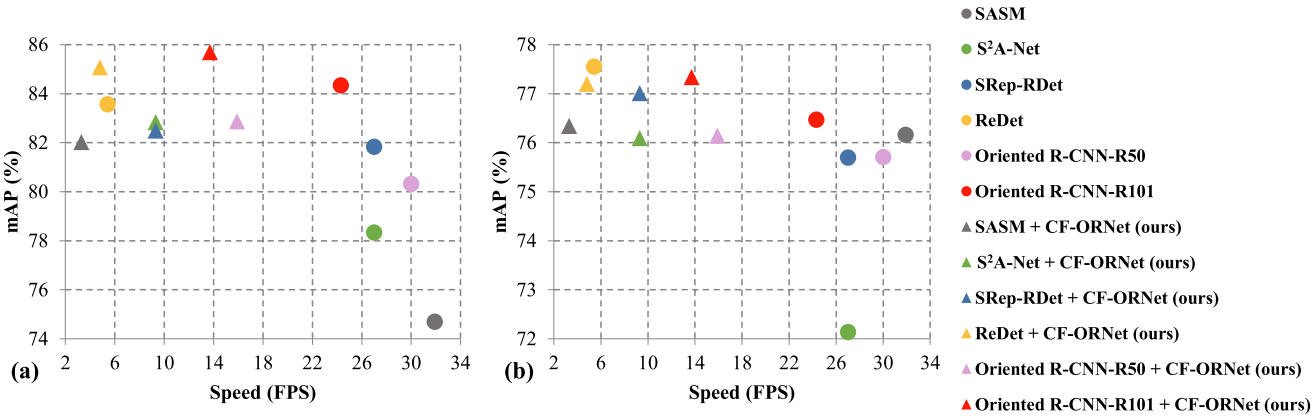


Fig. 9. Performance (mAP) of different methods on HRSC2016 and VEDAI datasets (circle represents other methods and triangle represents CF-ORNet method): (a) is the recognition result on the HRSC2016 dataset and (b) is the recognition result on the VEDAI dataset.

TABLE II
COMPARISON WITH OTHER METHODS ON HRSC2016 AND VEDAI DATASETS

Method	Backbone	mAP(%)		FPS
		HRSC2016	VEDAI	
SASM [46]	ResNet-50-FPN	74.70	76.16	**31.9**
S^{2}A-Net [47]	ResNet-50-FPN	77.73	72.14	27.0
SRep-RDet [29]	RepVGG-B1g2-FPN	81.23	75.70	27.0
ReDet [48]	ReResNet-50-ReFPN	83.31	**77.55**	5.4
Oriented R-CNN [11]	ResNet-50-FPN	80.32	75.71	30.0
Oriented R-CNN[11]	ResNet-101-FPN	**84.34**	76.47	24.3
SASM [46] + CF-ORNet (ours)	ResNet-50-FPN	82.02	76.34	3.3
S^{2}A-Net [47] + CF-ORNet (ours)	ResNet-50-FPN	82.83	76.09	9.3
SRep-RDet [29] + CF-ORNet (ours)	ResNet-50-FPN	82.49	77.01	9.3
ReDet [48] + CF-ORNet (ours)	ReResNet-50-FPN	85.07	77.20	4.8
Oriented R-CNN [11] + CF-ORNet (ours)	ResNet-50-FPN	82.87	76.14	**15.9**
Oriented R-CNN [11] + CF-ORNet (ours)	ResNet-101-FPN	**85.69**	**77.34**	13.7

and ResNet-50 as the backbone could get the fastest FPS (15.9). The satisfied FPS benefits from the speed advantage of oriented R-CNN, and the difficulty of category correction in CF-ORNet is greatly reduced due to the initial high-confidence objects. We can see that our method improves the feature discrimination ability and representation granularity.

F. Experiment V: Ablation Studies of Knowledge Distillation

To evaluate the efficiency of our lightweight model from knowledge distillation to filter grafting in fine-grained object recognition of HR-RSIs, we conduct ablation studies on the detector and CF-ORNet, respectively.

1) CWD for Detector: We design the CWD ablation experiment for the detector, as listed in Table III. We can see that distilled oriented R-CNN† (ResNet-50) has the highest FPS (30.0), which is 5.7 points of FPS improvement over oriented R-CNN (ResNet-101), an increase of 23%. In addition, student network has reduced 18.99M Params and 30.43G FLOPs compared with the teacher network. Since the structure of ResNet-50 is more compact than ResNet-101, the improvement of efficiency leads to the decrease of mAP (80.32%). After knowledge distillation, the mAP of the detector reaches 81.25% (an increase of 0.93%-point). Because knowledge distillation will not affect the structure of student, the recognition speed remains at the highest FPS (30.0). Furthermore, we combine the student networks before and after distillation with our CF-ORNet for ablation studies. Experiments show that compared with the teacher combination for oriented R-CNN (ResNet-101) + CF-ORNet, the distilled student compresses 19% of Params and has 15.9 FPS, improving recognition speed by 2.2-point. In terms of performance, oriented R-CNN (ResNet-50) + CF-ORNet obtain a mAP of 82.87% before knowledge distillation, and the performance gap with the teacher combination is 2.82%-point. After knowledge distillation, the performance of oriented R-CNN†(ResNet-50) + CF-ORNet increases by

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TABLE III
PERFORMANCE AFTER CWD

Method (Backbone)	Knowledge distillation	Params (M)	FLOPs (G MACs)	mAP (%)	FPS
Oriented R-CNN[11] (ResNet-101)	Teacher	60.14	121.51	**84.34**	24.3
Oriented R-CNN[11] (ResNet-50)	×	41.15	91.08	80.32	**30.0**
Oriented R-CNN† (ResNet-50)	√	41.15	91.08	81.25	**30.0**
Oriented R-CNN (ResNet-101)+CF-ORNet	—	101.24	129.36	**85.69**	13.7
Oriented R-CNN (ResNet-50)+CF-ORNet	×	82.25	98.93	82.87	**15.9**
Oriented R-CNN† (ResNet-50)+CF-ORNet	√	82.25	98.93	83.32	**15.9**

* † means the student network configured with knowledge distillation.

TABLE IV
PERFORMANCE AFTER WSLD

Method (Backbone)	Knowledge distillation	Params(M)	FLOPs(G MACs)	mAP(%)	FPS
Oriented R-CNN[11]	—	60.14	121.51	**84.34**	24.3
Oriented R-CNN†	—	41.15	91.08	81.25	**30.0**
Oriented R-CNN[11]+ CF-ORNet (ResNet-101)	—	101.24	129.36	**85.69**	13.7
Oriented R-CNN†+ CF-ORNet (ResNet-101)	×	82.25	98.93	83.32	15.9
Oriented R-CNN†+ CF-ORNet (ResNet-18)	×	52.34	92.90	81.94	**16.2**
Oriented R-CNN†+ CF-ORNet† (ResNet-18)	√	52.34	92.90	82.60	**16.2**

* † means the student network configured with knowledge distillation.

0.45% and reaches 83.32% mAP, and the performance gap with the teacher combination falls to 2.37%-point.

In general, the CWD method for detector effectively improves the performance of lightweight oriented R-CNN detector. Moreover, we introduce our CF-ORNet into distilled detector to achieve a lightweight model for fine-grained object recognition of HR-RSIs.

2) WSLD for CF-ORNet: We design a WSLD ablation experiment for CF-ORNet, as listed in Table IV. We compare the object recognition performance of CF-ORNet before and after distillation. The teacher detector (oriented R-CNN) and the student detector (oriented R-CNN†) are the baselines. Experiments show that oriented R-CNN†+ CF-ORNet (ResNet-18) achieves 81.94% mAP and 16.2 FPS before knowledge distillation, compared with oriented R-CNN†+ CF-ORNet (ResNet-101), a decrease of 1.38%-point. However, the lighter structure of ResNet-18 highly compresses the Params, thus reducing the Params from 82.25M to 52.34M (compression rate achieves 64%) and the FLOPs from 98.93G multiply-accumulate calculations (MACs) to 92.90G MACs. In addition, the backbone structure of ResNet-18 greatly speeds up the recognition, promoting the FPS from 15.9 to 16.2. In addition, we configure knowledge distillation to CF-ORNet with ResNet-18 to obtain an efficient structure of oriented R-CNN† + CF-ORNet†. Our method improves the recognition performance from 81.94% to 82.60% mAP (an increase of 0.66%-point).

The experiments prove that WSLD promotes the recognition performance of lightweight CF-ORNet. Yet, compared with the accuracy of the teacher combination (oriented R-CNN + CF-ORNet), there is still a gap, so it is indispensable to further explore ways to hoist performance.

TABLE V
PERFORMANCE AFTER FILTER GRAFTING

Method	Filter grafting		Params (M)	FLOPs (G MACs)	mAP (%)	FPS
	D	C				
Oriented R-CNN +CF-ORNet	×	×	101.24	129.36	**85.69**	13.7
Oriented R-CNN† +CF-ORNet†	×	×	52.34	92.90	82.60	**16.2**
Oriented R-CNN‡ +CF-ORNet†	√	×	52.34	92.90	83.94	**16.2**
Oriented R-CNN‡ +CF-ORNet‡	√	√	52.34	92.90	84.26	**16.2**

* † means the student network configured with knowledge distillation.
‡ means the student network configured with knowledge distillation and filter grafting, D is detector, C is classifier.

TABLE VI
PERFORMANCE UNDER DIFFERENT DEGRADATION VARIABILITIES ON THE HRSC2016 DATASET

Method	Original	Random brightness	Random blur	Random noise	Cloud shadow
Oriented R-CNN (baseline)	84.34	81.06	77.07	76.06	72.38
Oriented R-CNN +CF-ORNet (ours)	**85.69**	82.38	**77.46**	**77.18**	72.97
Oriented R-CNN‡ +CF-ORNet‡ (ours)	84.26	**83.34**	74.39	70.76	**73.36**

* ‡ means the student network and is configured with knowledge distillation and filter grafting. The indicators are mAP (%) under various degradation variabilities

G. Experiment VI: Ablation Studies of Filter Grafting

We also conduct ablation studies on filter grafting in knowledge distillation. As shown in Table V, the combinations involve the method before lightweight (oriented R-CNN + CF-ORNet), the method after lightweight (oriented R-CNN† + CF-ORNet†), the method after filter grafting configured for the detector (oriented R-CNN‡ + CF-ORNet†), and the method after filter grafting configured for both the detector and CF-ORNet (oriented R-CNN‡ + CF-ORNet‡). As the teacher combination, oriented R-CNN + CF-ORNet undoubtedly achieves the highest performance (85.69% mAP) and the slowest FPS (13.7). The method after lightweight (oriented R-CNN† + CF-ORNet†) compresses the Params from

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TABLE VII
COMPARISON WITH OTHER METHODS ON THE HRSC2016 DATASET

Method	Backbone	Params(M)	FLOPs (G MACs)	mAP(%)	FPS
R^3Det[28]	ResNet-50-FPN	42.07	132.62	57.87	22.3
Rotated Faster R-CNN[23]	ResNet-50-FPN	41.15	91.03	63.99	31.9
RoI Transformer[49]	ResNet-50-FPN	55.18	105.06	63.05	26.3
Rotated FCOS[50]	ResNet-50-FPN	31.94	81.00	73.63	**41.4**
SASM[46]	ResNet-50-FPN	36.61	75.97	74.70	31.9
S^2A-Net[47]	ResNet-50-FPN	38.63	77.52	77.73	27.0
SRep-RDet[29]	RepVGG-B1g2-FPN	59.68	124.05	81.23	27.0
ReDet[48]	ReResNet-50-ReFPN	31.69	48.39	83.31	5.4
Oriented R-CNN[11]	ResNet-50-FPN	41.15	91.08	80.32	30.0
Oriented R-CNN‡+CF-ORNet†(ours)	ResNet-34-FPN	33.37	35.46	78.44	23.4
Oriented R-CNN‡+CF-ORNet‡(ours)	ResNet-50-FPN	52.34	92.90	**84.26**	16.2

* ‡ means the student network and is configured with knowledge distillation and filter grafting.

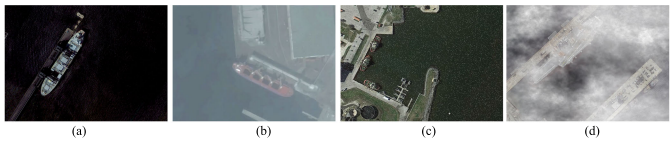


Fig. 10. Some examples of various image degradation models: (a) random brightness; (b) random blur, including Gaussian blur, average blur, and media blur; (c) random noise, including Gaussian noise, impulse noise, and salt-pepper noise; and (d) cloud shadow.

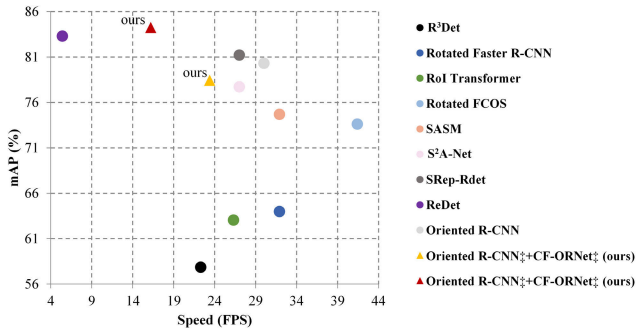


Fig. 11. Performance (mAP) of different methods on the HRSC2016 dataset (circle represents other methods and triangle represents CF-ORNet method).

101.24M to 52.34M, reduces the FLOPs from 129.36G MACs to 92.90G MACs, and improves the speed by 2.5 points to 16.2 FPS. However, lightweight leads to a serious decline in performance, down to 82.60% mAP. To further fill the performance gap between students and teachers, we introduce filter grafting into our model. The combination of oriented R-CNN‡+ CF-ORNet† achieves the 83.94% mAP, promoting 1.34%-point without any speed loss and structure changes. Then, we further configure the filter grafting on CF-ORNet in knowledge distillation to obtain the combination of oriented R-CNN‡+ CF-ORNet‡, which the performance is 84.26%, an increase of 0.32%-point mAP and narrows the gap with the teacher combination (3.09%-point–1.43%-point). At the same time, the number of Params is only 51% of that before lightweight, and FPS reaches 16.2 (an increase of 2.5-point).

To sum up, the experimental results demonstrate that our lightweight model from knowledge distillation to filter grafting achieves a better precision–speed balance (84.26% mAP–16.2 FPS). Parameter (52.34M) and FLOPs (92.90G MACs) show the compression ability and practical value.

H. Experiment VII: Performance Under Different Degradation Variabilities

Considering that real-world datasets often encounter image degradation and distortion in practical applications, we conducted performance experiments under different degradation variabilities in the HRSC2016 dataset, including random brightness, random blur, random noise, and cloud shadow. Fig. 10 shows some examples of different image degradations. Table VI lists the mAP comparison of oriented R-CNN, our model (oriented R-CNN + CF-ORNet), and our efficient model (oriented R-CNN‡+ CF-ORNet‡), which indicates that our models achieve SOTA performance in all degradation variabilities compared with the baseline model. The combination of oriented R-CNN + CF-ORNet is 0.39%-point and 1.12%-point higher than the baseline performance (mAP) in random blur and noise degradation, respectively. Our efficient model (oriented R-CNN‡+ CF-ORNet‡) has the advantage of 2.28%-point mAP over the baseline model with random brightness degradation and has the advantage of 0.98%-point mAP on the degraded images affected by cloud shadow.

The results demonstrate that our model has strong performance and robustness against various image degradation and distortion for fine-grained object recognition. It is worth noting that the lightweight model (oriented R-CNN‡+ CF-ORNet‡) shows powerful performance beyond the original model in the case of random brightness and cloud shadow due to the knowledge transfer conducted by the teacher networks through knowledge distillation and filter grafting.

I. Experiment VIII: Comparison Between Efficient Model and SOTA Methods

We evaluate the object recognition performance of our efficient model (oriented R-CNN‡+ CF-ORNet‡) with other

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SOTA methods on the HRSC2016 dataset. Table VII and Fig. 11 show the comparison results, in which the comparison methods include R^3Det [28], rotated faster R-CNN [23], RoI transformer [49], rotated fully convolutional one-stage (FCOS) [50], SASM [46], S^2A-Net [47], SRep-RDet [29], ReDet [48], and oriented R-CNN [11]. The results indicate that our efficient model (oriented R-CNN‡+ CF-ORNet‡) achieves the highest fine-grained object recognition performance (84.26% mAP) and gains satisfied 16.2-FPS speed, 52.34M Params, and FLOPs of 92.90G MACs. Compared with the oriented R-CNN using the same detector, our method has an advantage of 3.94%-point mAP, while the number of Params and FLOPs only increases by 11.19M and 1.82G MACs. The mAP has been highly improved in a nearly cost-free manner, demonstrating the powerful performance of our CF-ORNet. If the oriented R-CNN student network is replaced with a lighter backbone (ResNet-34), 78.44% mAP can be obtained. Although the performance is slightly reduced, it obtains less Params (33.37M) and the least FLOPs (35.46G MACs) compared with other methods, and its inference speed also has a breakthrough (23.4 FPS). Compared with other mainstream detectors, our efficient model (oriented R-CNN‡+ CF-ORNet‡) is 0.95%-point higher in performance (mAP) than that of ReDet (83.31%) while having an absolute advantage of 16.2 FPS over the speed of ReDet (5.4 FPS), which is three times that of the latter. Furthermore, our method has fewer Params (52.34M) than SRep-RDet (59.68M), but achieves a 3.03%-point mAP than SRep-RDet.

From the above results, our efficient model has advantages in recognition accuracy and model deployment compared with SOTA methods, and preferably balances the precision and speed of fine-grained object recognition in HR-RSIs.

V. Conclusion

In view of various scales, dense arrangement, highly similar objects, and high computational cost, we propose an efficient fine-grained object recognition method in HR-RSIs from knowledge distillation to filter grafting, which is composed of two parts, i.e., a two-stage fine-grained object recognition network including rotation object detection and CF-ORNet, a lightweight model of Part A from knowledge distillation to filter grafting. After extensive ablation studies and comparative experiments, the results show that our model achieves competitive performance and significantly outperforms most SOTA methods. What is more, it indicates that our coarse-to-fine recognition strategy to CF-ORNet can bring a performance improvement for most SOTA detectors, which shows the overall effectiveness. In particular, our lightweight model from knowledge distillation to filter grafting attains an expected performance to drop the deep model Params while placing them slightly within the proper performance range.

Although this work can balance the precision and speed of fine-grained object recognition with HR-RSIs to a certain extent, there is still room for improvement, such as finer feature mining and object extraction efficiency. In the future, we can introduce global–local feature fusion into CF-ORNet to improve the robust representation of fine-grained recognition, which is also extended to solve the problem of dense and small object recognition in HR-RSIs. In addition, for our lightweight model, we will explore more cost-effective grafting algorithms to facilitate training and save computational cost. Notably, few-shot fine-grained object recognition and robust fine-grained representations maybe push the future progression of the field. Furthermore, localization distillation and adaptive grafting could be a catalyst for the development of efficient object recognition models.

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